

Hossam BOUDRAA



+33 648 04 44 53 | hossam.boudraa@univ-amu.fr | [hossam-boudraa](https://hossam-boudraa.com)

Multilingual NLP researcher-engineer specializing in representation learning and large language models for information extraction, social and biomedical text.

Education

Master's Degree: Data Science and Artificial Intelligence

GALILÉE INSTITUTE – SORBONNE PARIS NORD UNIVERSITY

Paris - France
2021 – 2023

Master's Degree: Web Intelligence and Data Science (WISD)

SIDI MOHAMED BEN ABDELLAH UNIVERSITY

Fez - Morocco
2021 – 2023

Bachelor's Degree in Mathematics and Computer Science (SMI)

IBN ZOHR UNIVERSITY - AGADIR

Agadir - Morocco
2018 – 2021

Bachelor's Degree in French Studies

IBN ZOHR UNIVERSITY - AGADIR

Agadir - Morocco
2018 – 2021

Advanced Technician Diploma in Software Development

ISTA AGADIR

Agadir - Morocco
2016 – 2018

Technician Diploma in Computer Maintenance and Networking

ISTA AGADIR

Agadir - Morocco
2014 – 2016

Professional Experience

Research Engineer – SESSTIM, Aix-Marseille University

- Processed data from the French National Health Data System (SNDS) for a public health project.
- Developed prediction algorithms and a recommendation system based on AI techniques.
- Analyzed online forums to identify user profiles from unstructured data.

Marseille - France
November 2023 – Present

M2 Internship – LIS Laboratory, Aix-Marseille University

- Studied information transfer in French conversations.
- Evaluated pretrained language models (FlauBERT, CamemBERT).
- Generated automatic summaries from conversations.

Marseille - France
March – September 2023

L3 Internship – National School of Architecture of Agadir

- Designed a web application for grade management using PHP (Laravel).

Agadir - Morocco
January – June 2021

Observation Internship – Prefecture of Agadir

- Developed an application for managing rural associations using C#.

Agadir - Morocco
August – September 2019

Final-Year Internship – SoussNet Company

- Developed a stock management application in C++.

Agadir - Morocco
March – May 2018

Projects

EvalLLM2025 – Information Extraction Challenge

- PARTICIPATING IN THE EVALLLM2025 CHALLENGE FOCUSED ON INFORMATION EXTRACTION IN THE HEALTH DOMAIN USING FRENCH-LANGUAGE DOCUMENTS, IMPLEMENTING TWO MODEL-CENTRIC APPROACHES: A GENERATIVE MODEL LEVERAGING GPT4 AND AN AUTOENCODER-BASED MODEL UTILIZING BERT.

June 2025

NER Shared Task 2023 – Nested Named Entity Recognition in Arabic

- DEVELOPED A NAMED ENTITY RECOGNITION (NER) SYSTEM FOR ARABIC USING MARBERT AND ARABERT MODELS ON THE WOJOOD DATASET.

July 2023

SemEval 2023 – Multilingual Tweet Hostility Analysis

- DEVELOPED A MULTILINGUAL MODEL TO PREDICT HOSTILITY IN TWEETS, EVALUATED USING PEARSON'S R COEFFICIENT.

January 2023

iSarcasmEval – Sarcasm Detection in Arabic

- FINE-TUNED MARBERT WITH FOCAL LOSS FOR INTENTIONAL SARCASM DETECTION.

December 2022

NSFW Image Detection

- FLASK APPLICATION UTILIZING RESNET50 TO DETECT NSFW IMAGES.

October 2022

PageRank with Flask

- FLASK SERVICE TO COMPUTE PAGERANK SCORES FROM WEB GRAPHS.

December 2022

Facial Recognition – Golden Foot Players

- END-TO-END PIPELINE: SCRAPING, HAAR CASCADE, LBP, LBPH CLASSIFICATION, DJANGO DEPLOYMENT.

May 2022

Skills

Programming Languages	Python, R, SQL, C++, JavaScript, Bash
Deep Learning & NLP Frameworks	PyTorch, TensorFlow, HuggingFace Transformers, spaCy, NLTK, OpenCV
Model Architectures	Transformers, BERT,GPT-4, RoBERTa, Encoder-Decoder Models, Seq2Seq, Attention Mechanisms
Machine Learning Methods	Representation Learning, Zero-shot/Few-shot Learning, Tree Ensembles, Regression, Clustering, SVM
Modeling & Optimization	Unsupervised Learning, Risk Scoring, Causal Inference, Prompt Tuning, Transfer Learning
Bias & Social NLP:	LLM fine-tuning, Zero/Few-Shot Bias Probing, Multilingual NER, Hostility & Sarcasm Detection
Data Science & Visualization	Pandas, NumPy, Seaborn, Matplotlib, Plotly, Jupyter, Google Colab
Web & Backend Tools	Flask, FastAPI, Streamlit, Django, REST APIs, HTML/CSS, JSON, XML, Linux Scripting
Databases	MongoDB, MySQL, PostgreSQL, SQLite

Publications

Implicit Hate Target Span Identification in Zero and Few-Shot Settings with Selective Sub-Billion Parameter Models	
• HOSSAM BOUDRAA, BENOIT FAVRE, RAQUEL URENA	WOAH, ACL— July 2025
Leveraging Prompting for Information Extraction in Low-Resource French Medical Texts	
• HOSSAM BOUDRAA, ANAS BELFATHI	TALN— July 2025
Did You Get It? A Zero-Shot Approach to Locate Information Transfers in Conversations	
• ELIOT MAËS, HOSSAM BOUDRAA, PHILIPPE BLACHE, LEONOR BECERRA-BONACHE	LREC-COLING— May 2024

Languages

French:	Fluent (reading, speaking, writing)
English:	Professional proficiency (reading, speaking, writing)
Spanish:	Conversational proficiency (reading, speaking)
German:	Basics

Certifications

Deep Learning Specialization, Certification in deep learning covering neural networks, CNNs, and sequence models.	Coursera
Advanced SQL, Certification in advanced SQL for database manipulation.	HackerRank
Data Engineering, Skills in data pipelines, ETL, Airflow, and BigQuery.	IBM
Data Engineering, Retrieval-Augmented Generation with Embeddings & Vectors	Coursera

Teaching

Advanced Databases , Teaching Fellow (part-time), Master’s in Computer Science at Aix-Marseille University	Sept 2025- Dec 2025
Post-Relational Databases , Teaching Fellow (part-time), Master’s in Computer Science at Aix-Marseille University	Sept 2024- Dec 2024
Relational Database Systems , Teaching Fellow (part-time), 3rd-year students at Polytech Marseille	Sept 2024- Dec 2024

References

- Prof. My Abdelouahed Sabri , Professor, Department of Computer Science, abdelouahed.sabri@usmba.ac.ma
- Prof. MAHRAZ Mohamed Adnane , Professor, Department of Computer Science, mohamed.mahraz@usmba.ac.ma